

Phase 2 Logical Data Relationship Diagram

Official Website and Customer Enquiry Management System for a UK Auto Parts SME

Field	Details
Client Name Used	Auto Parts SME
Project Phase	Phase 2 Future Enhancement
Working Date	June 20XX
Prepared By	Business Analyst
Purpose	Define a logical data relationship view for enquiry and sales lead tracking before SQL and Power BI design.

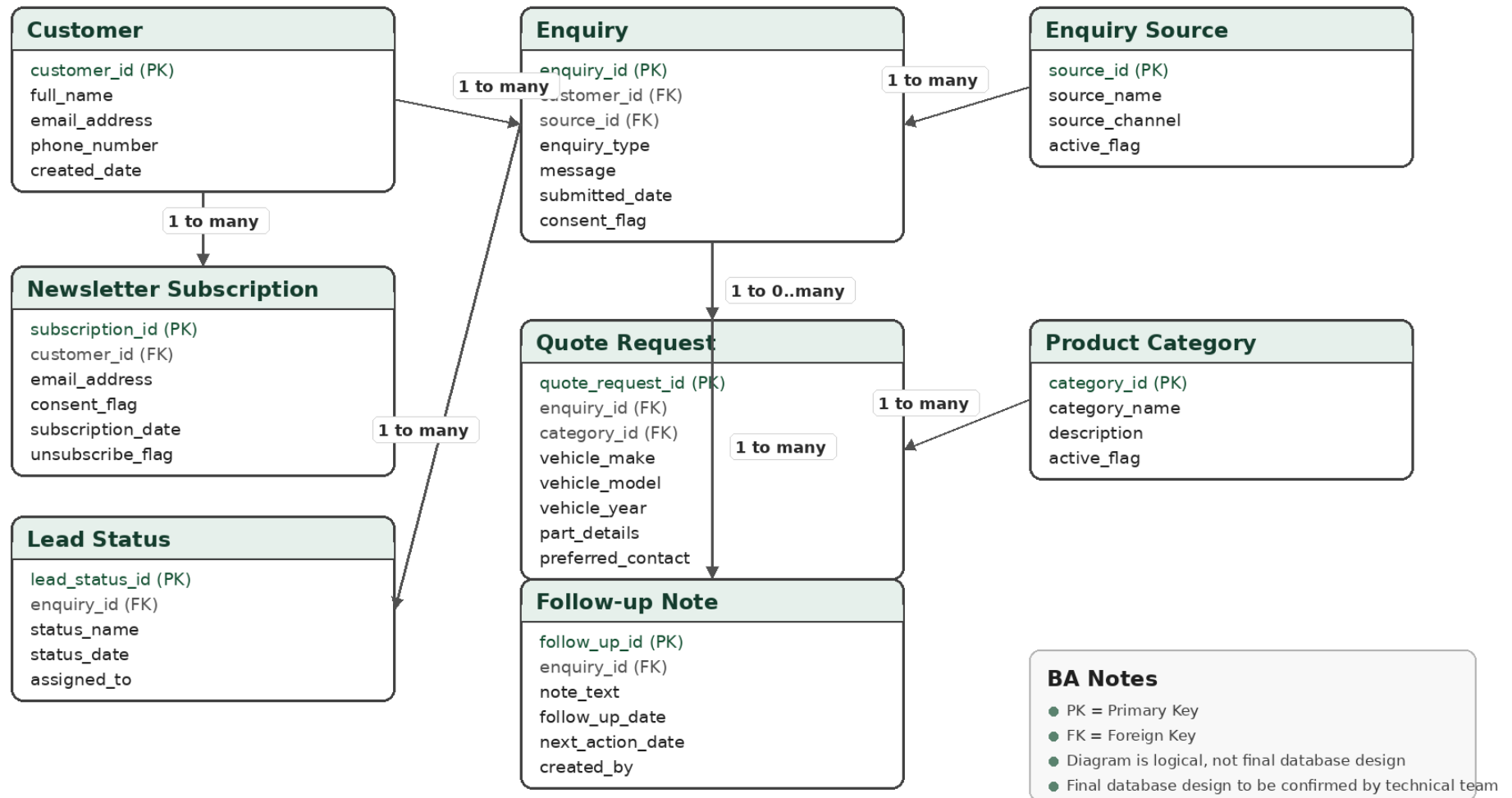
Note: This is a logical BA-level diagram. It is not a final physical database design. Final table structure, indexes, keys, data types, security rules, and retention rules should be confirmed by the technical team and relevant business/compliance owner.

1. Diagram

The diagram below shows the proposed logical relationship between customer enquiry data, quote requests, product categories, lead status, follow-up notes, enquiry source, and newsletter subscription data.

Phase 2 Logical Data Relationship Diagram

Customer Enquiry and Sales Lead Tracking Dashboard - SQL and Power BI Future Enhancement



2. Logical Entities

Entity	Business Purpose
Customer	Stores basic customer contact details captured through enquiry, quote, feedback, or newsletter forms.
Enquiry	Stores the main customer enquiry record submitted through the website.
Quote Request	Stores additional quote-specific details such as product category and vehicle details.
Product Category	Stores the agreed auto parts categories used in enquiry and reporting.
Lead Status	Stores lead progress such as New, Contacted, Quoted, Won, Lost, or Closed.
Follow-up Note	Stores business follow-up notes and next action dates for each enquiry.
Enquiry Source	Stores where the enquiry came from, such as website quote form, contact form, feedback form, or newsletter.
Newsletter Subscription	Stores newsletter consent and unsubscribe status.

3. Relationship Summary

Relationship	Meaning
Customer to Enquiry	One customer can submit many enquiries. Each enquiry should belong to one customer record where contact details are available.
Customer to Newsletter Subscription	One customer may have one or more newsletter subscription records depending on consent and unsubscribe history.
Enquiry Source to Enquiry	One enquiry source can be linked to many enquiries for reporting by source.
Enquiry to Quote Request	One enquiry may become one or more quote requests where the customer asks for parts pricing or availability response.
Product Category to Quote Request	One product category can be selected in many quote requests.
Enquiry to Lead Status	One enquiry can have multiple status updates during follow-up.
Enquiry to Follow-up Note	One enquiry can have many follow-up notes.

4. BA Validation Notes

Validation Area	BA Note
Data completeness	Customer name and email should be captured for quote and contact enquiries where required by the form.
Consent	Consent flag should be captured for forms where personal data is used to respond.
Lead status	Lead status values should be standardised to avoid reporting mismatch in Power BI.
Product category	Product category values should match the approved website categories.
Follow-up tracking	Follow-up notes should include follow-up date, next action date, and owner where required.
Reporting readiness	Fields should support Power BI metrics such as total enquiries, quote requests, top categories, pending follow-ups, conversion rate, and monthly trend.

5. Portfolio and Interview Note

Portfolio use: Upload the PNG diagram, editable Draw.io file, Word document, and PDF under the Phase 2 SQL and Power BI Reporting Enhancement section.

Interview explanation: For Phase 2, I prepared a logical data relationship diagram to show how customer, enquiry, quote request, product category, lead status, follow-up, source, and newsletter data would connect for future SQL and Power BI reporting. This helped translate business reporting needs into a structured data requirement view before technical database design.